

Chapter 3 Test

Kinematics

K	T	A	C
18/18	18/18	10/10	10/10

SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

Perfect!
Well Done!

THINKING

1. A football player kicks a football in a parabolic path. The football leaves the player's foot at an angle of 30.0° above the horizontal. The football lands 86.9 m horizontally from the point it was kicked 3.20 s later.

Show all your work

- a) What is the launch speed of the football?

find v_x , as it doesn't change

$$\vec{v}_x = \frac{\Delta \vec{x}}{\Delta t}$$

$$\vec{v}_x = \frac{86.9 \text{ m} [F]}{3.20 \text{ s}}$$

$$\vec{v}_x = 27.156 \text{ m/s} [F]$$

$$\vec{v}_i = \frac{27.156 \text{ m/s} [F]}{\cos 30^\circ}$$

$$\therefore \vec{v}_i = 31.357 \text{ m/s} [F 30^\circ \text{ up}]$$



$$v_x = 27.156 \text{ m/s} [F]$$

$$\cos 30^\circ = \frac{v_x}{v_i}$$

$$\cos 30^\circ = \frac{27.156}{v_i}$$

$$v_i = \frac{v_x}{\cos 30^\circ}$$

- b) What is the maximum height reached by the football?

$$\vec{v}_x = 27.156 \text{ m/s} [F]$$

$$\vec{v}_i = 31.357 \text{ m/s} [F 30^\circ \text{ up}]$$

$$\vec{v}_y = ?$$

$$\vec{v}_y = v_i \sin 30^\circ$$

$$\vec{v}_y = 31.357 \text{ m/s} \cdot \sin 30^\circ$$

$$\vec{v}_y = 15.679 \text{ m/s} [U]$$

$$\vec{v}_{y2} = 0 \text{ m/s} [U]$$

$$\vec{a} = -9.8 \text{ m/s}^2 [U]$$

$$\Delta d = ?$$

$$v_f^2 = v_i^2 + 2a\Delta d$$

$$v_f^2 - v_i^2 = 2a\Delta d$$

$$\Delta d = \frac{(0 \text{ m/s})^2 - (15.679 \text{ m/s})^2}{2(-9.8 \text{ m/s}^2)}$$

$$\therefore \Delta d = 12.542 \text{ m} [U]$$

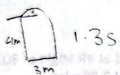
Final answer: $\vec{v}_i = 31.357 \text{ m/s} [F 30^\circ \text{ up}]$

Final answer: $\Delta d = 12.542 \text{ m} [U]$

2. A diver leaves the end of a 4.0 m high diving board and strikes the water 1.3 s later, 3.0 m beyond the end of the board. Considering the diver as a particle.
Show all your work

4 marks

- a) Calculate the initial velocity of the diver.



find v_x , doesn't change

$$v_x = \frac{\Delta x}{\Delta t}$$

$$v_x = \frac{3.0 \text{ m}}{1.3 \text{ s}}$$

$$v_x = 2.308 \text{ m/s [E]}$$

find v_y

$$v_{y1} = ?$$

$$\Delta d = -4 \text{ m [U]}$$

$$a = -9.8 \text{ m/s}^2 \text{ [U]}$$

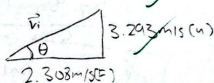
$$t = 1.3 \text{ s}$$

$$\Delta d_y = v_{y1} \Delta t + \frac{1}{2} a \Delta t^2$$

$$\Delta d_y = \frac{1}{2} a \Delta t^2$$

$$v_{y1} = \frac{-4 \text{ m [U]} - \frac{1}{2} (-9.8 \text{ m/s}^2) (1.3 \text{ s})^2}{1.3 \text{ s}}$$

$$v_{y1} = 3.293 \text{ m/s [U]}$$



$$v_i^2 = v_x^2 + v_y^2$$

$$v_i^2 = 2.308^2 + 3.293^2$$

$$v_i^2 = 16.17$$

$$v_i = 4.021 \text{ m/s}$$

$$\tan \theta = \frac{v_y}{v_x}$$

$$\theta = \tan^{-1} \left(\frac{v_y}{v_x} \right)$$

$$\theta = \tan^{-1} \left(\frac{3.293}{2.308} \right)$$

$$\theta = 54.97^\circ$$

$$\therefore v_i = 4.021 \text{ m/s [E } 54.97^\circ \text{ U]}$$

$$\text{Final answer: } v_i = 4.021 \text{ m/s [E } 54.97^\circ \text{ U]}$$

- b) Determine the velocity with which the diver enters the water.

v_x stays the same

$$v_{x1} = 3.293 \text{ m/s [U]}$$

$$a = -9.8 \text{ m/s}^2 \text{ [U]}$$

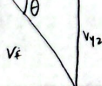
$$t = 1.3 \text{ s}$$

$$v_{y2} = v_{y1} + a \Delta t$$

$$v_{y2} = 3.293 \text{ m/s [U]} + (-9.8 \text{ m/s}^2) (1.3 \text{ s})$$

$$v_{y2} = -9.417 \text{ m/s [U]}$$

$$v_{x2} = 2.308 \text{ m/s [E]}$$



$$v_f^2 = v_{x2}^2 + v_{y2}^2$$

$$v_f^2 = 2.308^2 + (-9.417)^2$$

$$v_f^2 = 94.5727$$

$$v_f = 9.725 \text{ m/s}$$

$$\tan \theta = \frac{v_{y2}}{v_{x2}}$$

$$\theta = \tan^{-1} \left(\frac{v_{y2}}{v_{x2}} \right)$$

$$\theta = \tan^{-1} \left(\frac{-9.417}{2.308} \right)$$

$$\theta = 76.27^\circ$$

$$v_f = 9.725 \text{ m/s [E } 76.27^\circ \text{ D]}$$

$$\text{Final answer: } v_f = 9.725 \text{ m/s [E } 76.27^\circ \text{ D]}$$

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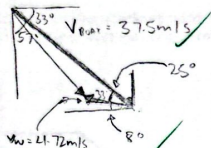
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3. A recreational boat is traveling at 135 km/h [S 57° E] relative to the water when the current's velocity is 17 km/h [W 8° N]. Calculate the velocity of the boat relative to an observer on the ground.

Show all your work

$$135 \text{ km/h} = 37.5 \text{ m/s}$$

$$17 \text{ km/h} = 4.72 \text{ m/s}$$



$$33^\circ - 8^\circ = 25^\circ$$

cos law

$$a = V_b = 37.5 \text{ m/s}$$

$$b = V_w = 4.72 \text{ m/s}$$

$$C = 25^\circ$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$c^2 = 37.5^2 + 4.72^2 - 2(37.5)(4.72) \cos 25^\circ$$

$$c^2 = 1167.62$$

$$c = 33.28 \text{ m/s}$$

sine law

$$c = 33.28 \text{ m/s}$$

$$C = 25^\circ$$

$$b = V_w = 4.72 \text{ m/s}$$

$$B = ?$$

$$\frac{\sin C}{c} = \frac{\sin B}{b}$$

$$\frac{\sin 25^\circ}{33.28} = \frac{\sin B}{4.72}$$

$$B = 3.44^\circ$$

$$33^\circ + 3.44^\circ = 36.44^\circ$$

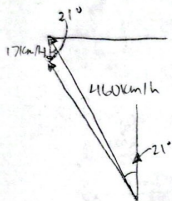
$$V_{\text{GH}} = 33.28 \text{ m/s [E } 36.44^\circ \text{ S]}$$

Final answer: 33.28 m/s [E 36.44° S]

4. A pilot is flying to a destination [N 21° W] of the present position. An air traffic controller on the ground tells the pilot that the wind velocity is 17 km/h [S]. The pilot knows that the airplane cruises at a speed of 460 km/h. Show all your work

[6 marks]

- a) Calculate the speed of the airplane relative to the ground.



$$a = 460 \text{ km/h}$$

$$b = 17 \text{ km/h}$$

$$C = 21^\circ$$

$$\cos 1a =$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$c^2 = 460^2 + 17^2 - 2(460)(17) \cos 21^\circ$$

$$c = 444.17 \text{ km/h}$$

Final answer: 444.17 km/h

- b) Determine the direction of the airplane.

Sine law

$$b = 17 \text{ km/h}$$

$$B = ?$$

$$C = 21^\circ$$

$$c = 444.17 \text{ km/h}$$

$$\frac{\sin B}{17} = \frac{\sin 21^\circ}{444.17}$$

$$B = 0.7859^\circ$$

$$21^\circ + 0.7859^\circ = 21.7859^\circ = \boxed{\text{N } 22^\circ \text{ W}}$$

Final answer: [N 22° W]

- c) What is the displacement of the airplane if the plane took 35 minutes to reach destination?

$$35 \text{ minutes} = 0.583 \text{ hrs}$$

$$d = vt$$

$$444.17 \text{ km/h} \times 0.583 \text{ hrs} = d$$

$$d = 259.099 \text{ km [N } 22^\circ \text{ W]}$$

Final answer: 259.1 km [N 22° W]

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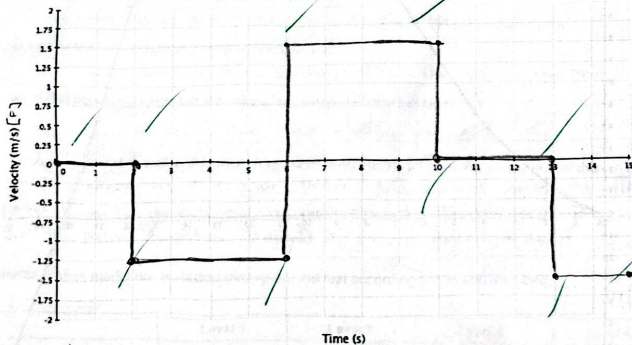
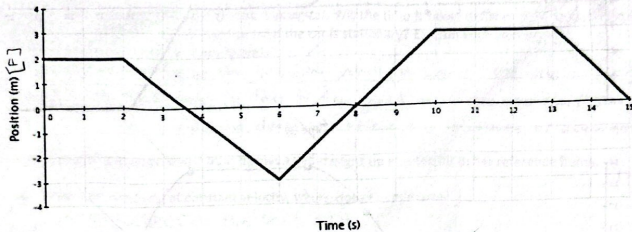
Chapter 1 Test Kinematics

K	T	A	C
/18	/18	10/10	10/10

SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

1. Convert the following position-time ($d-t$) graph into the corresponding velocity-time ($v-t$) graph.
Must show all your work including all calculations

[4 marks]



$$v = \frac{\Delta d}{\Delta t}$$

$$\textcircled{1} 0-2\text{ s} : \frac{2-2}{2\text{ s}} = 0\text{ m/s [F]}$$

$$\textcircled{2} 2-6\text{ s} : \frac{-3-2}{4\text{ s}} = -\frac{5}{4}\text{ m/s [F]}$$

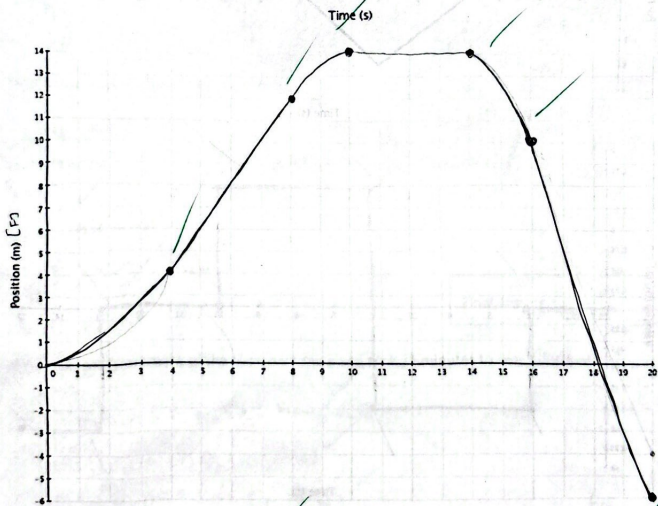
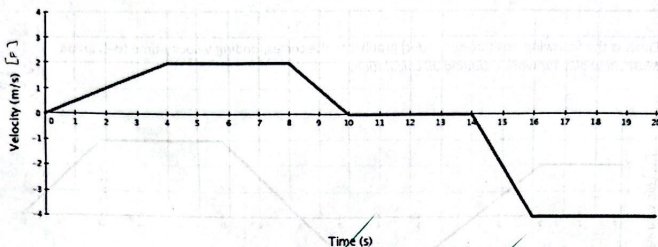
$$\textcircled{3} 6-10\text{ s} : \frac{3-(-2)}{4\text{ s}} = \frac{5}{4} = \frac{3}{2}\text{ m/s [F]}$$

$$\textcircled{4} 10-13\text{ s} : \frac{3-3}{3\text{ s}} = 0\text{ m/s [F]}$$

$$\textcircled{5} 13-15\text{ s} : \frac{0-3}{2\text{ s}} = -\frac{3}{2}\text{ m/s [F]}$$

2. Convert the following velocity-time ($v-t$) graph into the corresponding position-time ($d-t$) graph.
Must show all your work including all calculations

[6 marks]



① $0-4s: \frac{(0+2m/s) \cdot 4s}{2} = 4m [F]$
 ② $4-8s: \frac{(2+2m/s) \cdot 4s}{2} = 8m [F]$
 ③ $8-10s: \frac{(2+2m/s) \cdot 2s}{2} = 2m [F]$
 ④ $10-14s: \frac{(2+0m/s) \cdot 4s}{2} = 0m [F]$
 ⑤ $14-16s: \frac{(0+(-4m/s)) \cdot 2s}{2} = -4m [F]$
 ⑥ $16-20s: \frac{((-4)+(-4m/s)) \cdot 4s}{2} = -16m [F]$

Name: _____

Date: _____

COMMUNICATION

1. When observing the properties about the motion of a projectile object, describe two different/distinct variables that are constant?

In projectile motion, gravity is always the ^{positive} constant acceleration for the y-velocity. [2 marks]
 of objects, being 9.8 m/s^2 [Down].
 The x velocity is also always constant as the velocity is 0 m/s .

2. A ball is dropped from the window of a moving car. Will the time it takes to fall to the ground be the same, more, or less than the time it takes to fall if the car is stationary? Explain your answer.
 (Assume that air resistance is negligible).

If a ball were dropped from a moving vehicle, it would take the same amount of time to fall to the ground as if the car was stationary, because the x-velocity does not affect the y-velocity or acceleration. [2 marks]

3. A person sitting in an enclosed train throws a ball straight up into the air in her reference frame.

- a) If the train is moving at constant velocity, where does the ball land? [2 marks]

The ball lands in the same spot.

- b) If the train accelerates, where does the ball land?

The ball lands behind her, towards the rear of the train.

- c) If the train decelerates, where does the ball land?

The ball lands in front of her, towards the front of the train.

- d) If the train turns to the left around a curve, where does the ball land?

The ball lands to the right of the person.

4. An object is at rest as well as in motion at the same time. Provide an example to explain how this can be.

If two people are in a moving car, person A & B, and were viewed by person C standing still on the sidewalk, person A & B would seem to be in motion. But to person A, person B would be at rest relative to their viewpoint. [2 marks]

Communication marks are awarded throughout the test according to the shown rubric.

	Level 1	Level 2	Level 3	Level 4
Correct use of mathematical/ physical symbols, labels, and conventions (e.g. units, proper component form for the values, positive direction given, etc...)	Few mathematical/ physical conventions are used correctly. [1/2]	Some mathematical/ physical conventions are used correctly. [1]	Most mathematical/ physical conventions are used correctly. [1-1/2]	Consistently uses mathematical/ physical conventions correctly. [2]